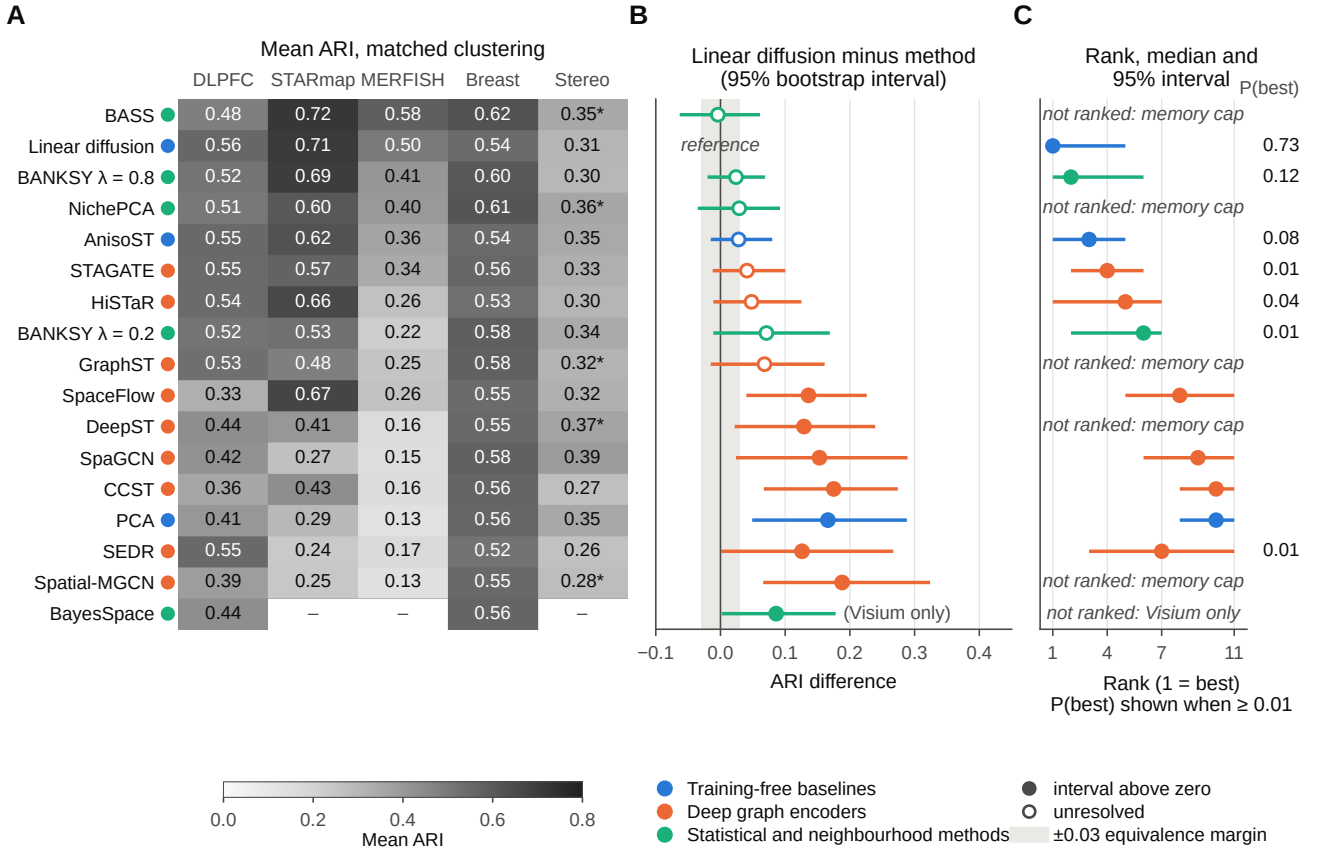

A
Mean ARI, matched clustering
DLPFC
STARmap
MERFISH
Breast
Stereo
BASS 0.48 0.72 0.58 0.62 0.35*
Linear diffusion 0.56 0.71 0.50 0.54 0.31
BANKSY λ = 0.8 0.52 0.69 0.41 0.60 0.30
NichePCA 0.51 0.60 0.40 0.61 0.36*
AnisoST 0.55 0.62 0.36 0.54 0.35
STAGATE 0.55 0.57 0.34 0.56 0.33
HiSTaR 0.54 0.66 0.26 0.53 0.30
BANKSY λ = 0.2 0.52 0.53 0.22 0.58 0.34
GraphST 0.53 0.48 0.25 0.58 0.32*
SpaceFlow 0.33 0.67 0.26 0.55 0.32
DeepST 0.44 0.41 0.16 0.55 0.37*
SpaGCN 0.42 0.27 0.15 0.58 0.39
CCST 0.36 0.43 0.16 0.56 0.27
PCA 0.41 0.29 0.13 0.56 0.35
SEDR 0.55 0.24 0.17 0.52 0.26
Spatial-MGCN 0.39 0.25 0.13 0.55 0.28*
BayesSpace 0.44 – – 0.56 –
0.0 0.2 0.4 0.6 0.8
Mean ARI
B
Linear diffusion minus method
(95% bootstrap interval)
reference
(Visium only)
−0.1 0.0 0.1 0.2 0.3 0.4
ARI difference
C
Rank, median and
95% interval
P(best)
not ranked: memory cap
0.73
0.12
not ranked: memory cap
0.08
0.01
0.04
0.01
not ranked: memory cap
not ranked: memory cap
0.01
not ranked: memory cap
not ranked: Visium only
1 4 7 11
Rank (1 = best)
P(best) shown when ≥ 0.01
Training-free baselines
Deep graph encoders
Statistical and neighbourhood methods
interval above zero
unresolved
±0.03 equivalence margin